

Tianyu Kong

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Research Interests

Mathematical modeling · Numerical analysis · Partial differential equations · Multiscale analysis · Mathematical physics · Quantum materials

Academic Position

William H. Kruskal Instructor, University of Chicago, Chicago, IL 08.2026 – Present
Committee on Computational and Applied Mathematics (CCAM)

Education

University of Minnesota Twin Cities, Minneapolis, MN 09.2021 – 05.2026
PhD in Applied Mathematics

- Advisor: Mitchell Luskin, Alexander B. Watson
- Dissertation: *Multiscale Modeling of Electron Dynamics in Twisted Bilayer Graphene*

University of Chicago, Chicago, IL 09.2017 – 06.2021
Honors BS in Applied Mathematics & Physics

Honors and Scholarships

SIAM Student Travel Award	09.2025
Great Lakes SIAM Travel Award	08.2025
UMN Doctoral Dissertation Fellowship	05.2024
SIAM Student Travel Award	03.2024
UMN Vanky Men Memorial Fellowship	12.2023
UChicago Dean's List	07.2018, 07.2019

Publications

1. **Higher-order continuum models for twisted bilayer graphene**
Solomon Quinn, **Tianyu Kong**, Mitchell Luskin, Alexander B. Watson
Journal of Mathematical Physics 66.10 (2025)
2. **Interacting twisted bilayer graphene with systematic modeling of structural relaxation**
Tianyu Kong, Alexander B. Watson, Mitchell Luskin, Kevin D. Stubbs
Electronic Structure. 035001 (2025).
3. **Modeling of electronic dynamics in twisted bilayer graphene**
Tianyu Kong, Diyi Liu, Mitchell Luskin, Alexander B. Watson
SIAM Journal on Applied Mathematics. 84, 1011 (2024).
4. **Bistritzer-MacDonald dynamics in twisted bilayer graphene**
Alexander B. Watson, **Tianyu Kong**, Allan H. MacDonald, Mitchell Luskin
Journal of Mathematical Physics 64.3 (2023). *Editor's Choice*

Research Experience

- Student Assistant**, Lawrence Berkeley National Laboratory *05.2024 – 08.2024*
Advisor: Chao Yang and Lin Lin *05.2023 – 08.2023*
- Modeled electron-electron interactions, uniform strain, and structural relaxation in magic-angle twisted bilayer graphene.
 - Implemented Hartree-Fock (HF) and Coupled Cluster (CC) methods using the PySCF framework in Python.
- Undergraduate Research Assistant**, University of Chicago *04.2020 – 05.2021*
Advisor: Mary Silber
- Implemented optimization algorithms (steepest descent, quasi-Newton) in MATLAB to calculate Minimum Action Paths for noise-induced transitions in Lorenz systems.

Teaching Experience

- Teaching assistant**, University of Minnesota Twin Cities *09.2021 – 05.2024*
- *Honors Calculus*
 - *Calculus for College of Science and Engineering I - II*
- Grader**, University of Minnesota Twin Cities *09.2022 – 05.2024*
- *Mathematical Modeling and Applied Mathematics I - II (Graduate level)*
 - *Functional Analysis (Graduate level)*
- Grader**, University of Chicago *09.2018 – 05.2020*
- *Mathematical Methods for Physical Sciences*
 - *Honors Calculus I - III*

Talks and Seminar Presentations

- Continuum Model for Relaxed Moire Bilayer Graphene** *11.2025*
SIAM Conference on Analysis of Partial Differential Equations, Pittsburgh, PA
- Multiscale Modeling of Electron Dynamics in Twisted Bilayer Graphene** *10.2025*
University of Minnesota Duluth Math Colloquium, Duluth, MN
- Multiscale Modeling of Electron Dynamics in Twisted Bilayer Graphene** *10.2025*
UMN Seminar of Mathematical Applications & Computations, Minneapolis, MN
- Multiscale Modeling in Twisted Bilayer Graphene with Strain and Relaxation** *09.2025*
SIAM Great Lakes Section Annual Meeting, Chicago, IL
- “Magic” in Moiré Materials – An Applied Mathematics Perspective** *03.2025*
UMN Seminar of Mathematical Applications & Computations, Minneapolis, MN
- Modeling of Electronic Dynamics in Twisted Bilayer Graphene** *07.2024*
International workshop on 2D and moiré materials, Roscoff, France
- Modeling of Electronic Dynamics in Twisted Bilayer Graphene** *05.2024*
SIAM Conference on Materials Science, Pittsburgh, PA
- Modeling of Electronic Dynamics in Twisted Bilayer Graphene** *03.2024*
Brin Mathematics Research Center Workshop, College Park, MD

Poster Presentations

- Continuum model and interactions for relaxed twisted bilayer graphene** *06.2026*
IMSI Mathematical Aspects of 2D Quantum Materials and Meta-materials, Chicago, IL

Interacting Twisted Bilayer Graphene with Systematic Structural Relaxation	<i>05.2025</i>
Simons Foundation Moiré Materials Magic Workshop, New York, NY	
A Comparison of Minimum Action Methods for Computing Noise-induced Transitions of the Lorenz System	<i>05.2021</i>
SIAM Conference on Applications of Dynamical Systems (DS21), Remote	

Academic Services

Co-organizer (with Guillaume Bal), SIAM Great Lakes Meeting, Chicago, IL	<i>09.2025</i>
◦ Co-organized minisymposium <i>Computation and analysis for quantum materials and wave phenomenon</i>	
Webmaster , SIAM student chapter, University of Minnesota	<i>09.2024 – 05.2026</i>
◦ Managed the website and official documents for the SIAM student chapter at UMN	
◦ Co-organized the annual student Integration Bee contest for undergraduate students	

Technologies

Python, MATLAB, Julia, R, MySQL